CS560 - UI/UX Design & Implementation

7-2 Short Paper: Analyzing Usability Study Findings

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Analyzing Usability Study Findings

Usability studies help designers understand how users interact with digital products and identify areas where improvements are needed. The study of *User Interface Factors of Mobile UX: A Study with an Incident Reporting Application* examined how different mobile interface designs affected usability and user experience when participants completed incident reporting tasks. The researchers compared three application prototypes to determine how the sequence of interface elements and navigation methods influenced user performance and satisfaction. The findings provide valuable insights into how mobile workflows can be designed to support efficient and positive user experiences.

Workflow Description

The primary workflow in the study involved completing and submitting an incident report using a mobile application. Users were required to enter information about an environmental observation or incident through a series of form fields. This process included reviewing information, entering details, navigating through the form, and submitting a completed report. Although the reporting task remained the same across all prototypes, the organization of the interface elements differed. The original mobile user interface presented information using a traditional layout. The first redesigned prototype reorganized the sequence of form elements to create a more logical flow for users completing the reporting process. The second redesigned prototype used a similar sequence but changed the navigation style by incorporating tabs instead of relying entirely on scrolling. The purpose of these workflow variations was to determine whether users could complete reporting tasks more efficiently and with greater satisfaction when interface elements were arranged differently.

User Challenges and Interaction Issues

The study identified several challenges that affected the user's experience. Users had trouble when interface elements were not presented in a sequence that matched their natural thought process. When users had to switch between unrelated information fields or search for the next required step, they experienced hesitation and additional cognitive effort. These workflow interruptions made the reporting process feel less intuitive and reduced overall satisfaction. The researchers reported that the original interface received lower usability ratings than both redesigned prototypes. The System Usability Scale (SUS) score for the original interface was significantly lower than the scores received by the redesigned versions. Similarly, the User Experience Questionnaire (UEQ) results showed higher pragmatic quality scores for the redesigned interfaces, indicating that users found those workflows easier to understand and complete. These results suggest that the arrangement of interface elements had a meaningful impact on usability and user satisfaction. The study found a slight difference between tab-based navigation and scrolling when users completed short forms. This finding suggests users could navigate efficiently using either method if the information were organized logically. Rather than the navigation style itself, users were more influenced by the overall structure and sequence of the workflow.

Evaluation of the Study

Overall, the study effectively analyzed and explained the usability challenges experienced by participants. One strength of the research is its use of controlled experimentation. By changing only specific interface factors while keeping the task consistent, the researchers were able to isolate the impact of workflow design decisions on user experience. This approach strengthened the credibility of the findings and allowed for more meaningful comparisons between prototypes. Another strength was the use of established usability metrics, including the System Usability Scale (SUS) and the User Experience Questionnaire (UEQ). These tools are widely used in usability research and provide reliable measures of both usability and user satisfaction (Brooke, 1996; Schrepp et al., 2017). The study also included task completion time data, allowing the researchers to evaluate both objective and subjective aspects of performance. One limitation of the study is that it relied heavily on quantitative measurements. While the numerical data clearly demonstrated differences between the interfaces, additional qualitative observations or participant comments would have provided deeper insight into why users preferred certain workflows. According to Tullis and Albert (2023), combining performance metrics with qualitative feedback often produces a more complete understanding of usability problems because it reveals both what users do and why they behave that way.

Priorities for Improvement

If I were improving an existing application based on this study, my priority would be optimizing the sequence and organization of interface elements. The research clearly showed that workflow structure had a stronger impact on usability than navigation style. Users should be guided through tasks in a logical order that matches their expectations and reduces cognitive effort. When information is presented naturally, users can focus on completing their goals rather than figuring out how the interface works. My second priority would be reducing unnecessary navigation and simplifying data-entry processes. Mobile users often complete tasks in environments where distractions are common. Streamlined workflows can reduce frustration and improve task completion rates. Clear labels, grouped information, and consistent navigation patterns would further support usability and efficiency. Finally, I would prioritize ongoing usability testing throughout the design process. User expectations and behaviors change over time, making continuous evaluation essential for maintaining a positive user experience. Regular testing can identify workflow problems early and ensure design decisions are aligned with user needs.

Conclusion

The study demonstrates that workflow organizations play a critical role in mobile usability. Although navigation methods such as tabs and scrolling had a negligible effect on short forms, the sequence of interface elements significantly influenced user satisfaction and perceived usability. The researchers effectively used established usability metrics to support their conclusions and provide actionable design recommendations. These findings highlight the importance of designing workflows that align with users' natural thought processes and support efficient task completion. Focusing on logical organization, simplified navigation, and continuous usability testing, designers can create mobile applications that provide more intuitive and satisfying user experiences.

No AI-generated tools are used.

References:

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